
DUET: Principled Experience Replay for LLM Agent Reinforcement Learning

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Abstract

Experience replay from strong teacher agents is an appealing way to overcome the cold-start problem in on-policy RL for LLM agents. Yet in GRPO-style updates, inserting replayed teacher trajectories into the same rollout batch as student trajectories is not a benign form of supervision: it changes both the advantage estimates and the distributional assumptions underlying the policy gradient. We identify two systematic biases induced by this teacher-mixing replay. First, successful teacher rollouts contaminate the group baseline, suppressing the advantages assigned to successful on-policy student rollouts and thereby discouraging exploration. Second, teacher trajectories are off-policy, but exact teacher-student importance correction can be infeasible when teacher likelihoods are unavailable, leaving standard GRPO updates with an uncorrected distribution mismatch. Therefore, we propose **DUET**, a principled experience-replay framework that first corrects these biases through baseline separation and density-ratio correction, and then extracts complementary teacher signals through two channels: token-level behavior cloning in the action channel and potential-based state-progress shaping in the state channel. Together, these components let DUET exploit teacher trajectories beyond direct replay while adaptively attenuating teacher influence as the student improves, turning teacher data into complementary signals for action imitation and state-guided exploration. Across ALFWorld and WebShop with two model scales, DUET achieves the best performance in all used settings, improving the strongest prior baseline by 13.0 percentage points on average and by 17.5 points in the weakest-agent regime. Code is available at <https://anonymous.4open.science/r/DUET-DC26>.

1 Introduction

Large language model (LLM) agents are increasingly trained with reinforcement learning (RL) to solve interactive tasks that require multi-turn decision making in external environments [Shridhar et al., 2020, Yao et al., 2022, 2023]. Group-based policy-gradient methods such as GRPO [Shao et al., 2024] have become a practical backbone for this training. However, these methods remain difficult to use in sparse-reward settings. When a weak student agent rarely completes a task on its own, the on-policy rollout batch contains little positive signal, and the policy gradient has almost nothing to reinforce.

A natural way to address this cold-start problem is to replay successful trajectories from a stronger teacher model. Recent methods follow this idea with promising results. LUFFY [Yan et al., 2025] mixes teacher demonstrations into GRPO rollout groups, while CHORD [Zhang et al., 2026] combines teacher-mixed RL with a weighted imitation objective. These methods show that teacher data can substantially accelerate LLM-agent training. Most existing uses of teacher replay, however,

37 treat the teacher trajectory mainly as an action sequence: the teachers outputs either enter the policy-
38 gradient batch or serve as supervised targets.

39 This view leaves part of the teacher trajectory unused. In interactive tasks, a successful trajectory
40 contains not only the teachers outputs, but also the sequence of environment states that mark progress
41 toward completion. These state-level signals can guide exploration without requiring the student to
42 imitate the teachers exact actions, which is useful when multiple reasoning traces or action sequences
43 can solve the same task.

44 At the same time, even teacher experience replay is not statistically benign in GRPO-style training.
45 When teacher and student rollouts are normalized together, two problems arise. First, successful
46 teacher trajectories can inflate the group reward statistics, reducing the advantages assigned to the
47 few successful on-policy rollouts (**baseline contamination**). This directly weakens the exploratory
48 successes that RL should amplify. Second, teacher trajectories are off-policy, creating a **distribution**
49 **mismatch**. In principle, their gradients should be corrected by a teacher-student importance ratio,
50 but exact correction is often unavailable when the teacher is closed-source or its tokenizer does not
51 align with the student’s. Naive replay can therefore keep teacher gradients influential even after
52 the student no longer benefits from copying them. Both problems are most severe in the cold-start
53 regime, where teacher replay is most needed.

54 We propose **DUET**, a teacher-replay framework that uses teacher trajectories as two complemen-
55 tary sources of signal. The *Action Channel* determines how teacher outputs affect the policy up-
56 date. It separates teacher and on-policy baselines so that teacher rewards do not distort on-policy
57 credit assignment, and uses a discriminator-based density-ratio weight to approximate the missing
58 teacher-student correction when exact likelihood ratios are unavailable. The same weight reduces
59 teacher-action gradients as the student improves, so teacher influence is controlled by the estimated
60 distribution gap rather than by a fixed training-step schedule. In parallel, the *State Channel* uses
61 teacher observations to build a progress map over environment states. This map provides dense re-
62 ward shaping for on-policy rollouts, guiding exploration toward teacher-like progress without requir-
63 ing the student to copy teacher actions. Because the State Channel shapes only on-policy rollouts
64 while the Action Channel corrects teacher-action gradients, DUET can use both signals together
65 without reintroducing the baseline contamination caused by naive teacher mixing.

66 We evaluate DUET on ALFWorld and WebShop using 1.5B and 3B LLMs as students, together
67 with a fixed trajectory experience collected from a 72B teacher model. Through experiments, DUET
68 achieves strong success rate compared to other baselines. The gains are largest for the weaker 1.5B
69 students, where cold start is most severe: DUET improves over the strongest prior baseline by 17.5
70 percentage points on both ALFWorld and WebShop. Ablations show that baseline separation is
71 essential for stable training, while density-ratio correction and the State Channel provide comple-
72 mentary gains.

73 Our contributions are:

- 74 1. We identify two failure modes of direct teacher mixing in GRPO: baseline contamination
75 from high-reward teacher rollouts, and uncorrected distribution mismatch when teacher
76 likelihoods are unavailable.
- 77 2. We propose DUET, a dual-channel teacher-replay framework that uses teacher actions for
78 corrected action-level learning and teacher states for progress-based reward shaping.
- 79 3. We show that DUET improves LLM-agent RL across two interactive environments and two
80 student model scales, with the largest gains in the cold-start regime.

81 2 Related Work

82 **Group-based RL for interactive LLM agents.** LLM agents are increasingly trained in multi-
83 turn, sparse-reward environments [Shridhar et al., 2020, Yao et al., 2022, 2023]. GRPO [Shao et al.,
84 2024], building on PPO [Schulman et al., 2017] and RLHF [Ouyang et al., 2022], replaces a learned
85 critic with group-relative rewards and has become a practical backbone. Recent agent-RL work adds
86 unified environments [Xi et al., 2024], stability analysis [Wang et al., 2025], and finer-grained credit
87 assignment [Feng et al., 2025]. DUET is complementary to these on-policy advances: it studies how
88 cached teacher trajectories should enter a GRPO-style update without corrupting on-policy credit.

Teacher replay beyond imitation. Demonstrations have long been used for cold-start exploration in imitation, demonstration-augmented, and offline-to-online RL [Ross et al., 2011, Hester et al., 2018, Nair et al., 2020, Kostrikov et al., 2022]. For LLM post-training, LUFFY mixes off-policy reasoning traces with on-policy GRPO rollouts [Yan et al., 2025], while CHORD reframes SFT as a dynamically weighted auxiliary objective inside on-policy RL [Zhang et al., 2026]. These methods show that expert data helps weak learners, but mainly use teacher trajectories as action or token sequences. DUET instead treats demonstrations as structured interaction experience: teacher actions provide a corrected replay signal, while teacher states provide progress guidance.

Correcting off-policy teacher actions. Direct teacher mixing is not benign supervision inside group-normalized policy gradients. Successful teacher rollouts can shift the group baseline, and teacher actions are off-policy. Classical off-policy policy gradients therefore motivate behavior-target correction [Degris et al., 2012]. When exact teacher likelihoods are unavailable, density-ratio estimation is a standard surrogate [Sugiyama et al., 2012]; binary classifiers estimate distributional differences in generative modeling and adversarial imitation [Goodfellow et al., 2014, Ho and Ermon, 2016]. DUET applies this principle narrowly: a discriminator-based replay weight approximates teacher-student correction, while baseline separation prevents teacher rewards from suppressing on-policy successes.

State progress and reward shaping. Teacher trajectories also contain environment states, not only output actions. In ALFWorld and WebShop, intermediate observations can reveal progress toward household or product-search goals [Shridhar et al., 2020, Yao et al., 2022]. Reward shaping offers a natural lens: potential-based shaping preserves optimal policies under $F = \gamma\Phi(s') - \Phi(s)$ [Ng et al., 1999]. DUET uses a coarser empirical progress potential built from successful teacher state visitation. The bonus is applied only to on-policy rollouts, keeping state guidance separate from corrected teacher-action replay.

3 DUET: Dual-Channel Teacher Utilization

3.1 Setup

We train a student policy $\pi_\theta(a \mid s)$ with GRPO [Shao et al., 2024] in sparse-reward interactive environments. For each task, the student samples n_o on-policy rollouts G^o . We also use a fixed cache of successful trajectories $\{\tau^\beta\}$ collected from a stronger teacher policy π_β . Following teacher-mixed replay [Yan et al., 2025, Zhang et al., 2026], each training group includes n_β teacher trajectories G^β :

$$G = G^o \cup G^\beta, \quad |G| = n_o + n_\beta. \quad (1)$$

Our goal is to use this cache without letting off-policy teacher trajectories distort the group-normalized policy-gradient update.

3.2 Overview

DUET uses teacher trajectories through two complementary channels. The **Action Channel** determines how teacher actions enter the policy update: it separates teacher and on-policy baselines, estimates a teacher-student density ratio when exact importance correction is unavailable, and adds behavior cloning as a cold-start bridge. The **State Channel** uses teacher observations to estimate task progress and shape rewards for on-policy rollouts. Thus, teacher actions provide corrected action-level supervision, while teacher states provide progress-based exploration guidance.

3.3 Action Channel

Baseline contamination. In teacher-mixed GRPO, successful teacher trajectories can raise the group baseline and suppress credit assigned to on-policy rollouts. Let p_o be the on-policy success rate in a group, and assume replayed teacher trajectories are successful, $R^\beta = 1$. The pooled reward mean is

$$\mu_{\text{pool}} = \frac{n_o p_o + n_\beta}{n_o + n_\beta}. \quad (2)$$

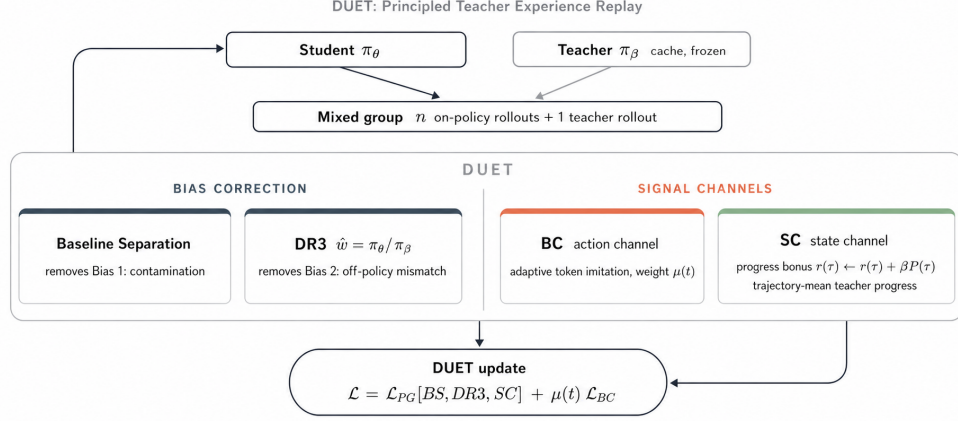


Figure 1: **DUET overview.** Teacher trajectories provide both action-level and state-level information. The Action Channel uses teacher actions with baseline separation and density-ratio correction. The State Channel uses teacher observations to construct a progress map and shape rewards for on-policy rollouts.

Under this pooled baseline, the expected centered reward of an on-policy rollout is

$$\mathbb{E}_{i \in G^o} [R^{o,(i)} - \mu_{\text{pool}}] = p_o - \frac{n_o p_o + n_\beta}{n_o + n_\beta} = -\frac{n_\beta}{n_o + n_\beta} (1 - p_o). \quad (3)$$

Thus, whenever $p_o < 1$, pooled normalization introduces a negative shift for on-policy samples. The shift is largest during cold start, but persists until the student succeeds on nearly all rollouts. It therefore suppresses the rare exploratory successes that RL should amplify.

Asymmetric baseline separation. DUET removes this shift by centering on-policy samples only against the on-policy subgroup:

$$\hat{A}^{o,(i)} = \frac{R^{o,(i)} - \mu^o}{\sigma^o}, \quad \mu^o = \frac{1}{|G^o|} \sum_{j \in G^o} R^{o,(j)}, \quad (4)$$

where σ^o is the standard deviation of on-policy rewards. Teacher samples are centered against the pooled mean and scaled by the same on-policy standard deviation:

$$\hat{A}^{\beta,(i)} = \frac{R^{\beta,(i)} - \mu_{\text{pool}}}{\sigma^o}. \quad (5)$$

This asymmetric rule keeps on-policy advantages centered within G^o , while giving successful teacher samples positive advantage when the student is weak:

$$R^{\beta,(i)} - \mu_{\text{pool}} = 1 - \mu_{\text{pool}} = \frac{n_o}{n_o + n_\beta} (1 - p_o). \quad (6)$$

As the student improves, the teacher advantage naturally decreases. Using σ^o also avoids the zero-variance issue of success-filtered teacher batches.

Density-Ratio Repair (DR3). We refer to the missing teacher–student correction as a *repair* on the standard policy ratio, and abbreviate the resulting estimator as DR3. Teacher actions are generated by π_β , while the standard GRPO ratio

$$\rho_t = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \quad (7)$$

only accounts for the change from the old student policy to the current one. When teacher log-probabilities are unavailable, DUET estimates the missing teacher–student ratio with a discriminator $D_\phi(s, a)$ trained to distinguish on-policy from teacher state–action pairs:

$$\hat{w}(s, a) = \frac{D_\phi(s, a)}{1 - D_\phi(s, a)} \approx \frac{\pi_\theta(a | s)}{\pi_\beta(a | s)}. \quad (8)$$

151 D_ϕ is a lightweight MLP over sequence-level features and outputs the probability that (s, a) comes
 152 from the on-policy student rather than the teacher cache. It is updated jointly with the actor at every
 153 policy step from a rolling buffer of recent on-policy and teacher samples, so $\hat{w}$ tracks the evolving
 154 student distribution rather than a fixed snapshot. The feature set and training procedure are detailed
 155 in Appendix A.1. This gives the corrected teacher replay term

$$\mathcal{L}_{\text{PG}}^{\text{tch}}(\theta) = -\mathbb{E}_{(s,a) \sim G^\beta} \left[\hat{A}^\beta \cdot \text{clip}(\hat{w}(s, a)\rho_t, 1 - \varepsilon, 1 + \varepsilon) \right]. \quad (9)$$

156 We use $\hat{w}$ as a bias-mitigating replay weight rather than an exact likelihood ratio, since the teacher
 157 cache is success-filtered. It also attenuates teacher gradients according to the estimated teacher-
 158 student distribution gap, instead of a fixed training-step schedule.

159 **Adaptive behavior cloning.** Early in training, the discriminator has limited data, so DUET adds
 160 token-level behavior cloning on teacher trajectories:

$$\mathcal{L}_{\text{BC}}(\theta) = -\mathbb{E}_{(s,a) \sim G^\beta} [\log \pi_\theta(a \mid s)]. \quad (10)$$

161 Its coefficient $\mu(t)$ is high during cold start and decays as the discriminator becomes informative.
 162 This provides dense action supervision before the density-ratio correction stabilizes. Implementation
 163 details are in Appendix A.

164 3.4 State Channel

165 The State Channel extracts progress information from states visited by successful teacher trajectories.
 166 For each teacher trajectory $\tau^\beta = (s_0^\beta, \dots, s_{T_\tau^\beta}^\beta)$, we assign every observation a within-trajectory
 167 progress value $\phi(s_t^\beta; \tau^\beta) = t/T_\tau^\beta$. We then aggregate these values across the teacher cache into a
 168 state potential

$$\Phi(s) = \frac{1}{|\mathcal{T}_\beta(s)|} \sum_{(\tau^\beta, t) \in \mathcal{T}_\beta(s)} \phi(s_t^\beta; \tau^\beta), \quad \Phi : \mathcal{S} \rightarrow [0, 1], \quad (11)$$

169 where $\mathcal{T}_\beta(s) = \{(\tau^\beta, t) : s_t^\beta = s\}$ is the set of all teacher visits to state s , and unmatched states are
 170 assigned $\Phi(s) = 0$. For an on-policy trajectory $\tau = (s_0, a_0, \dots, s_T)$, DUET computes

$$P(\tau) = \frac{1}{T+1} \sum_{t=0}^T \Phi(s_t), \quad r'(\tau) = r(\tau) + \lambda P(\tau). \quad (12)$$

171 Here $r(\tau)$ is the environment task reward and $\lambda > 0$ is the shaping coefficient (chosen as a small
 172 constant to avoid overwhelming the task reward). This trajectory-level shaping gives dense feedback
 173 when environment rewards are sparse. It is inspired by potential-based reward shaping [Ng et al.,
 174 1999], but we use an empirical trajectory-level score and do not claim exact policy invariance. The
 175 bonus is applied only to on-policy rollouts; teacher trajectories keep their original rewards.

176 3.5 Combined objective

177 The DUET loss combines shaped on-policy RL, corrected teacher replay, and cold-start behavior
 178 cloning:

$$\mathcal{L}_{\text{DUET}}(\theta) = \sum_{i \in G^o} \mathcal{L}_{\text{PG}}^{\text{on}}(\hat{A}_{\text{shaped}}^{o,(i)}; \theta) + \sum_{j \in G^\beta} \mathcal{L}_{\text{PG}}^{\text{tch}}(\hat{A}^{\beta,(j)}, \hat{w}^{(j)}; \theta) + \mu(t) \mathcal{L}_{\text{BC}}(\theta). \quad (13)$$

179 $\hat{A}_{\text{shaped}}^o$ is obtained by substituting the shaped reward $r'(\tau)$ from Eq. (12) for $R^{o,(i)}$ in the on-policy
 180 advantage formula Eq. (4); teacher samples use $\hat{A}^\beta$ from Eq. (5) together with the density-ratio
 181 weight $\hat{w}$ from Eq. (8).

182 4 Experiments

183 4.1 Setup

184 **Environments.** We evaluate DUET on two interactive LLM-agent benchmarks through the
 185 AgentGym-based evaluation toolkit [Xi et al., 2024]. **ALFWorld** [Shridhar et al., 2020] is a

Table 1: **Main results.** Test-set success rate on ALFWorld and WebShop under a fixed 100-step training budget. Best result in each column is shown in bold; the strongest non-DUET baseline is underlined. The Δ row reports DUET’s gain over the strongest non-DUET baseline.

Method	1.5B-ALFWorld	1.5B-WebShop	3B-ALFWorld	3B-WebShop
OnPolicy GRPO	1.0%	0.5%	47.0%	2.0%
LUFFY	5.5%	5.5%	61.5%	38.0%
CHORD	27.0%	11.5%	67.0%	39.0%
SFT+GRPO	<u>30.0%</u>	<u>18.5%</u>	59.5%	24.0%
DUET (Ours)	47.5%	36.0%	77.5%	45.5%
Δ vs. strongest baseline	+17.5pp	+17.5pp	+10.5pp	+6.5pp

text-based household instruction-following environment with long action horizons and sparse goal-completion rewards. **WebShop** [Yao et al., 2022] is a simulated e-commerce navigation environment with structured product search, attribute matching, and partial-credit rewards. For both environments, we report test-set success rate using the same evaluation protocol for all methods.

Models. We use Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct as student policies where teacher-replay is highly beneficial. The teacher data is a fixed offline trajectory cache collected from Qwen2.5-72B-Instruct and filtered to successful executions. The same teacher cache is used for all methods that require teacher trajectories.

Baselines. We compare against four baselines: (i) **OnPolicy GRPO** [Shao et al., 2024], which uses no teacher data; (ii) **LUFFY** [Yan et al., 2025], which mixes teacher trajectories into GRPO with a policy-shaping term; (iii) **CHORD** [Zhang et al., 2026], which adds a manually scheduled supervised objective on top of teacher-mixed GRPO; and (iv) **SFT+GRPO**, which first trains on teacher data with supervised learning and then continues with on-policy GRPO. Implementations of LUFFY and CHORD follow the authors’ released code where available.

Training protocol. All methods use the same GRPO backbone, batch size, group size, optimizer, KL coefficient, teacher cache, and total training budget. We train each method for 100 update steps, using a fixed training budget rather than tuning the horizon separately for each method. For teacher-replay methods, each group contains both on-policy student rollouts and replayed teacher trajectories under the same mixing ratio. Additional implementation details are provided in Appendix A.

4.2 Main results

Table 1 reports test-set success rate after the fixed 100-step training budget. DUET achieves the highest success rate among all reproduced baselines in all four model–environment settings, improving over the strongest non-DUET baseline by an average of 13.0 percentage points.

DUET helps most in the cold-start regime. The gains are largest for the weaker 1.5B students, where on-policy GRPO receives little useful reward signal. On 1.5B, DUET improves over the strongest baseline by 17.5 percentage points on both ALFWorld and WebShop. The gains remain positive for the 3B students, but the margins are smaller, consistent with stronger students suffering less from cold start.

Teacher mixing alone is not enough. LUFFY improves over purely on-policy GRPO in most settings, confirming that teacher trajectories are useful. However, directly mixing teacher data does not address baseline contamination or off-policy mismatch. CHORD obtains stronger cold-start performance by adding supervised learning on teacher trajectories, but its teacher influence is controlled by a predefined schedule. DUET improves over these baselines by combining corrected teacher-action replay with progress-based reward shaping from teacher states.

Gain magnitude follows the cold-start pattern. Equation (3) suggests that baseline contamination is strongest when the on-policy success rate p_o is small. The empirical pattern is consistent with this prediction: DUET’s gain over the strongest non-DUET baseline is +17.5pp on both 1.5B

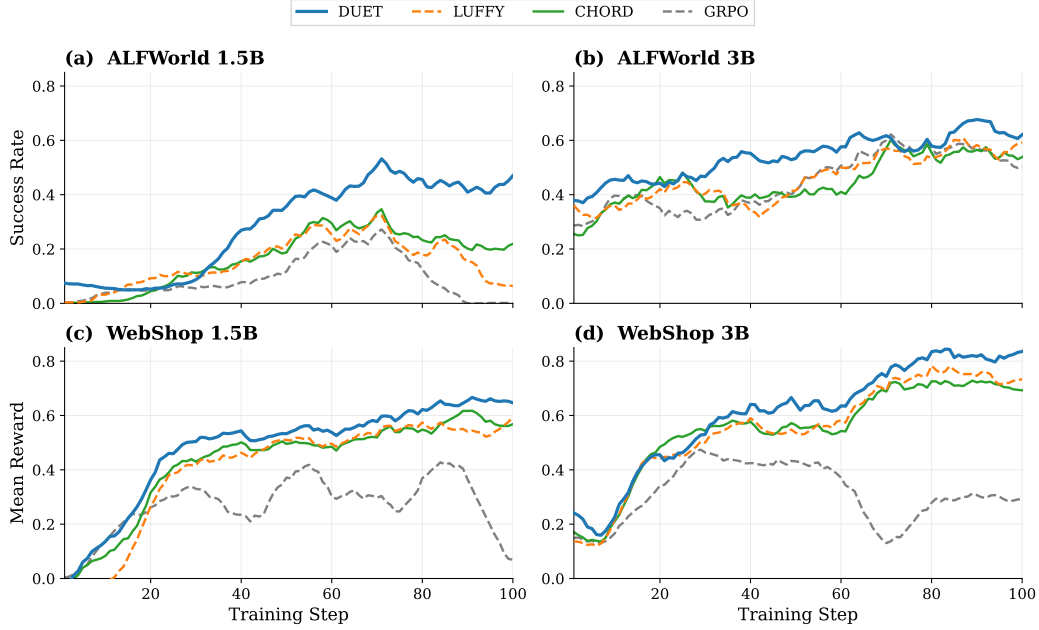


Figure 2: **Learning curves under the fixed 100-step budget.** The y-axis reports on-policy success rate on ALFWorld and mean reward on WebShop. DUET improves final performance in all four model–environment settings, with the clearest gains in the 1.5B cold-start regime.

Table 2: **Component ablations on 1.5B students.** Each row removes one DUET component while keeping all others fixed. The rightmost column reports the average drop from full DUET across the two environments.

Variant	Component removed	1.5B-AF	1.5B-WS	Avg. Δ
DUET (full)	–	47.5%	36.0%	–
w/o baseline sep.	Advantage normalization	0.0%*	0.0%*	–41.8pp
w/o DR3 weighting	Density-ratio correction	38.5%	9.5%	–17.8pp
w/o State Channel	Progress reward shaping	31.0%	1.0%	–25.8pp
w/o BC	Cold-start imitation	34.0%	16.5%	–16.5pp
OnPolicy GRPO	No teacher data	1.0%	0.5%	–

* Training collapses to 0% within the first ~ 50 steps when baseline separation is removed.

settings, where on-policy GRPO nearly collapses, and decreases to +10.5pp / +6.5pp on the 3B settings, where the on-policy stream already provides useful signal. This pattern supports the view that DUET is most valuable when teacher replay is needed but direct teacher mixing is most biased.

Learning dynamics. Figure 2 shows training curves under the fixed 100-step budget. DUET attains the strongest final performance in all four panels and also shows more stable improvements, with the clearest separation in the 1.5B settings. On ALFWorld, the curves report on-policy success rate. On WebShop, they report mean reward to reflect the environment’s partial-credit feedback during training.

4.3 Component ablations

Table 2 ablates DUET’s components on the 1.5B settings, where cold start makes the differences most visible. Each row removes one component while keeping the rest of the configuration fixed.

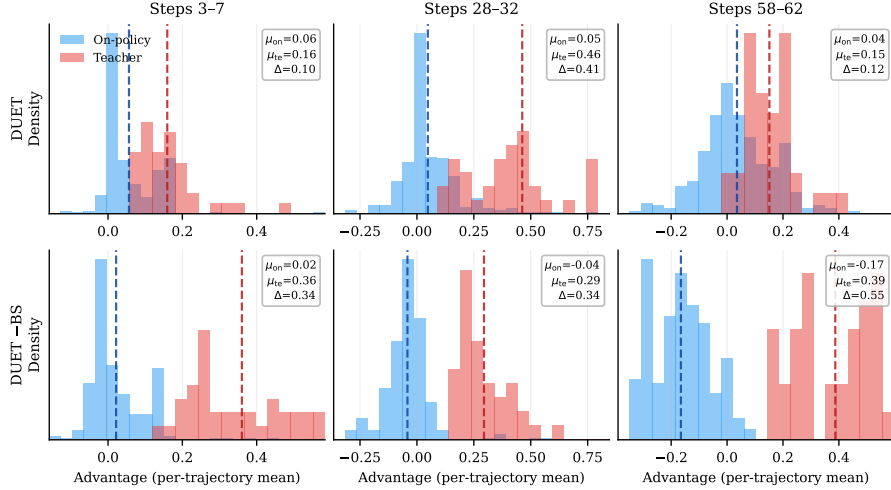


Figure 3: **Advantage distributions with and without baseline separation** on 1.5B-ALFWorld. Top row: full DUET; on-policy means stay near zero throughout training. Bottom row: DUET with baseline separation removed; on-policy means drift negative as teacher rewards inflate the shared GRPO baseline, in the direction predicted by Eq. (3).

Baseline separation is critical. Removing baseline separation collapses both 1.5B settings to 0.0%, making it the largest ablation effect. This matches the failure mode predicted by Eq. (3): without separation, high-reward teacher trajectories shift the baseline upward and suppress credit for rare successful on-policy rollouts.

Visualizing the contamination bias. Figure 3 shows per-trajectory advantage distributions on 1.5B-ALFWorld at three training stages, with and without baseline separation. With baseline separation, the on-policy mean stays near zero throughout training ($\mu_{on} = 0.06, 0.05, 0.04$). Without it, the on-policy distribution drifts negative as teacher rewards raise the shared baseline ($\mu_{on} = 0.02 \rightarrow -0.04 \rightarrow -0.17$). This matches the sign and direction predicted by Eq. (3), and provides a direct visual explanation for the collapse observed in Table 2.

Different components address different failure modes. Removing the State Channel reduces performance by 16.5pp on ALFWorld and 35.0pp on WebShop, showing that progress-based shaping provides substantial exploration guidance. Removing DR3 causes a significant drop on both environments, suggesting that distribution correction matters most when the teacher–student gap persists over longer or more diverse interactions. Behavior cloning also contributes in both environments by providing dense cold-start supervision before the discriminator-based correction stabilizes.

4.4 Action Channel dynamics

The Action Channel is designed to reduce teacher influence according to training signals rather than a fixed step schedule. Figure 4 reports the quantities that control this behavior. The effective teacher replay weight decreases over training, the BC coefficient decays from its cold-start value, and discriminator accuracy rises above chance. Together, these trends indicate that DUET relies most strongly on teacher actions early in training and gradually attenuates their effect.

Two complementary fade-out paths. DUET reduces teacher influence through two signals. The first comes from asymmetric baseline separation: by Eq. (5), the teacher centered reward is proportional to $(1 - p_o)$, so it shrinks as on-policy success improves. The second comes from DR3: $\hat{w}$ tracks the teacher–student distribution gap and decreases as teacher samples become easier to distinguish from on-policy samples. These mechanisms respond to reward progress and distributional progress, respectively, allowing teacher influence to recede without a fixed step-indexed schedule.

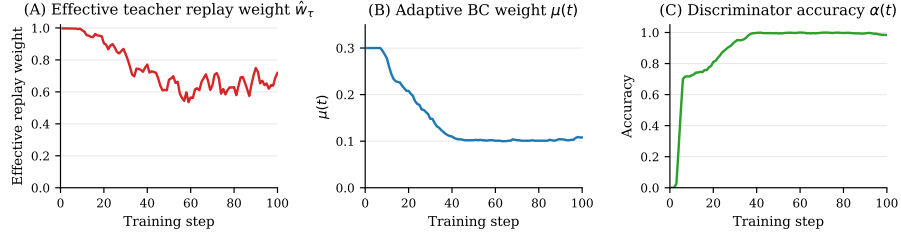


Figure 4: **Self-attenuation in the Action Channel.** (A) The effective teacher replay weight $\hat{w}_\tau$ decreases over training. (B) The adaptive behavior-cloning coefficient $\mu(t)$ decays after the cold-start phase. (C) Discriminator accuracy $\alpha(t)$ rises above chance, providing the signal used to attenuate teacher-action influence.

5 Discussion and Limitations

When DUET helps most. DUET is designed for the regime where teacher replay is most useful but also most likely to bias the update: sparse-reward training with weak student policies. In this regime, the on-policy reward stream contains few successful rollouts, so teacher trajectories provide necessary guidance. At the same time, directly mixing high-reward teacher rollouts into GRPO can distort the group baseline and over-emphasize off-policy actions. The empirical results follow this pattern: DUET gives its largest gains on the 1.5B students, where on-policy GRPO nearly collapses, and smaller but still positive gains on the 3B students, where the on-policy stream is already more informative.

Which components are load-bearing. The ablations suggest that the components play different roles. Baseline separation is the most consistently important mechanism: removing it collapses both 1.5B settings, matching the baseline-contamination failure mode analyzed in Section 3.3. The State Channel provides dense exploration guidance and is especially helpful when task progress can be inferred from intermediate observations. DR3 is more environment-dependent. It has a large effect on WebShop, where the teacher–student mismatch persists over longer and more diverse interactions, but a smaller marginal effect on ALFWorld, where baseline separation and cold-start imitation already close much of the gap. This suggests that DUET should be viewed as a framework for separating teacher signals, rather than as a claim that every component contributes equally in every environment.

Dependence on teacher data. DUET assumes access to a cache of successful teacher trajectories. The quality, diversity, and filtering of this cache can affect all three sources of teacher influence: the action replay signal, the density-ratio estimate, and the state-progress map. A narrow teacher cache may bias the State Channel toward a limited set of solution paths, while a noisy or weak teacher cache may provide unreliable progress signals. Our experiments use successful trajectories from a stronger teacher model, which matches the intended cold-start setting but does not cover cases where teacher demonstrations are suboptimal, heterogeneous, or adversarially filtered.

Limits of the current density-ratio correction. Our DR3 implementation estimates a trajectory-level replay weight and applies it uniformly to the teacher tokens in that trajectory. This is a coarse approximation. A token-level or step-level density ratio could focus the correction on teacher actions that are unlikely under the current student, and might reduce the need for a separate behavior-cloning bridge. We leave this finer-grained correction to future work.

Limits of the State Channel. The State Channel relies on matching on-policy observations to states seen in successful teacher trajectories. This works naturally in environments where progress can be recognized from textual observations or structured attributes, but it may be less reliable when observations are highly aliased, noisy, or only weakly correlated with task completion. In such settings, a learned progress model or uncertainty-aware shaping rule may be needed.

Scope of evaluation. Our evaluation focuses on two representative interactive LLM-agent settings: embodied text interaction and web navigation. These settings capture sparse rewards, multi-turn

interaction, and teacher-guided cold start, which are the main conditions DUET is designed for. The magnitude of the gains may vary with horizon length, teacher quality, and how much progress information is exposed in the environment state.

6 Conclusion

We studied teacher replay for reinforcement learning of LLM agents under GRPO-style training. Although teacher trajectories provide valuable cold-start signal, directly mixing them with on-policy rollouts can bias group-relative advantage estimation and leave teacher-action gradients uncorrected when teacher likelihoods are unavailable. DUET addresses these issues with two complementary uses of teacher data: corrected action-level replay through baseline separation and density-ratio weighting, and state-level guidance through progress-based reward shaping. Experiments on ALF-World and WebShop show that DUET improves over teacher-replay and imitation-based baselines across student model scales, especially when the student has little initial on-policy success. We hope this perspective encourages future work to treat teacher demonstrations as structured interaction traces, rather than only as action sequences to imitate.

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369 A Implementation Details

370 This appendix provides implementation details for DUET beyond the main method description. We
371 focus on algorithmic choices needed for reproducibility, and omit code-level configuration names.

372 A.1 Density-ratio discriminator

373 DUET estimates the teacher–student density ratio with a lightweight discriminator. The discrimina-
374 tor is a three-layer MLP with hidden width 64 and tanh activations. Its input consists of sequence-
375 level features computed from each rollout, including summary statistics of token log-probabilities,
376 low-probability tail ratios, KL-to-reference statistics, and response length. When high-dimensional
377 hidden features are used, we first apply a LayerNorm projection with dropout before concatenating
378 them with scalar features.

379 The discriminator is trained with binary cross-entropy to distinguish on-policy samples from teacher-
380 replay samples. Training batches are class-balanced between the two sources, and we use label
381 smoothing with coefficient 0.1. The optimizer is Adam with learning rate 3×10^{-4} and weight
382 decay 0.005. Discriminator examples are drawn from a rolling buffer of recent on-policy and teacher
383 samples. To emphasize recent policy behavior while retaining stability, older samples are down-
384 weighted exponentially by age and the weights are normalized to preserve the loss scale.

385 We delay applying the density-ratio weights until the discriminator has accumulated a sufficient
386 buffer of examples. Before this warmup completes, teacher samples are trained without DR3 weight-
387 ing and behavior cloning remains at its cold-start strength. This avoids applying unreliable density-
388 ratio estimates during the first few actor updates.

389 For inference, we use a smoothed copy of the discriminator parameters to compute $\hat{w}$. This reduces
390 short-term feedback between discriminator updates and the actor loss that consumes the resulting
391 weights. We also apply a temperature to discriminator logits before converting them into density
392 ratios, which prevents over-confident early predictions from producing unstable replay weights.

393 A.2 Replay-weight clipping and stabilization

394 The raw density-ratio estimate

$$\hat{w}(s, a) = \frac{D_\phi(s, a)}{1 - D_\phi(s, a)}$$

can be noisy early in training. We therefore clip the effective teacher replay multiplier used in the actor loss. The clipping threshold is adapted to keep the effective sample size of teacher weights within a stable range. Concretely, we maintain a dual variable that increases when the effective sample size falls below a target fraction of the teacher batch and decreases otherwise. The resulting upper bound prevents a small number of teacher samples from dominating the policy-gradient update.

In distributed training, discriminator features and labels are synchronized across workers before discriminator updates. The discriminator parameters and scalar statistics used by the actor loss are also synchronized across workers. This ensures that all ranks apply consistent replay weights and behavior-cloning coefficients.

A.3 Adaptive behavior-cloning coefficient

The behavior-cloning coefficient $\mu(t)$ is driven by discriminator accuracy rather than by a fixed step-indexed schedule. Let $\bar{d}_t$ be an exponential moving average of discriminator accuracy, and let d_{floor} denote the accuracy threshold below which the discriminator is treated as uninformative. We map discriminator accuracy to the BC coefficient with

$$g_t = \text{clip}\left(\frac{1 - \bar{d}_t}{1 - d_{\text{floor}}}, 0, 1\right), \quad \mu(t) = \mu_{\min} + (\mu_{\max} - \mu_{\min})g_t. \quad (14)$$

Thus, BC remains strong while the discriminator is near chance and decays as the discriminator becomes informative. During discriminator warmup, we treat accuracy as chance-level so that $\mu(t)$ stays at $\mu_{\max}$.

We use token-level cross-entropy on teacher trajectories as the BC objective. No additional token-level weighting is applied inside the BC loss, so $\mu(t)$ is the only coefficient controlling the strength of behavior cloning. The hyperparameters used in the 1.5B experiments are shown in Table 3.

Table 3: **Adaptive BC hyperparameters.** $\mu_{\max}$ and $\mu_{\min}$ are the maximum and minimum BC coefficients. d_{floor} is the discriminator-accuracy threshold used in Eq. (14).

Setting	$\mu_{\max}$	$\mu_{\min}$	d_{floor}	EMA decay
1.5B-ALFWorld	0.30	0.05	0.40	0.50
1.5B-WebShop	0.30	0.10	0.60	0.20

A.4 State Channel implementation

The State Channel builds a progress potential Φ from successful teacher trajectories. For each teacher trajectory, observations are assigned normalized progress values according to their relative position in the trajectory. These values are aggregated across the teacher cache to form a map from environment states to progress scores in $[0, 1]$. States that cannot be matched to the teacher cache receive progress value 0.

State matching is environment dependent. For ALFWorld, observations are represented with a compact state key derived from the text observation. For WebShop, the state key additionally uses product and attribute information, since progress depends on matching user-specified constraints such as product type, brand, price, and other attributes. This makes the progress map more robust to surface-form variation in WebShop observations.

For an on-policy trajectory $\tau = (s_0, a_0, \dots, s_T)$, the progress score is the mean potential over visited states:

$$P(\tau) = \frac{1}{T+1} \sum_{t=0}^T \Phi(s_t). \quad (15)$$

We use $\lambda = 0.20$ for the shaping coefficient and add the bonus $\lambda P(\tau)$ to the trajectory reward. The bonus is applied only to on-policy rollouts. Teacher trajectories keep their original success reward, since adding progress bonuses to already-successful teacher samples would increase their advantages and partially counteract DR3.

432 The State Channel bonus is added to the on-policy reward used by the policy-gradient update, but
 433 it is not used to shift the teacher baseline. This keeps state-level shaping separate from the Action
 434 Channel correction.

435 A.5 Asymmetric baseline separation

436 DUET uses the asymmetric baseline rule described in Section 3.3. For on-policy samples, the reward
 437 baseline is the mean reward of the on-policy subgroup:

$$\hat{A}^{o,(i)} = \frac{R^{o,(i)} - \mu^o}{\sigma^o}. \quad (16)$$

438 For teacher samples, the baseline is the pooled group mean:

$$\hat{A}^{\beta,(i)} = \frac{R^{\beta,(i)} - \mu_{\text{pool}}}{\sigma^o}. \quad (17)$$

439 Both subgroups are scaled by the on-policy standard deviation σ^o . This choice avoids the zero-
 440 variance problem caused by success-filtered teacher batches, where all teacher rewards are often
 441 identical. It also ensures that teacher rewards do not shift the on-policy baseline, while teacher
 442 samples still receive positive advantage when the student is weak.

443 A.6 Training system

444 Training uses a distributed PPO-style RL loop with fully sharded data-parallel policy training. Roll-
 445 out generation is served asynchronously, and environment interaction is parallelized across sepa-
 446 rate actor pools. The student policy, rollout workers, discriminator, and reward computation are
 447 synchronized once per actor update. All methods in the main experiments use the same training
 448 infrastructure, teacher cache, optimization budget, and evaluation protocol.

449 B Extended main results: reward mean

450 Table 1 reports strict success rate (score ≥ 1.0) for compactness. WebShop also supports partial
 451 credit; we report reward mean on the cell where the two metrics are most distinct: 1.5B-WebShop.
 452 ALFWorld is binary, so its reward mean coincides with strict success rate.

Method (1.5B-WebShop)	Strict SR	Reward mean
OnPolicy GRPO	0.5%	0.152
LUFFY	5.5%	0.573
CHORD	11.5%	0.603
SFT + GRPO	<u>18.5%</u>	<u>0.641</u>
DUET (Ours)	36.0%	0.706

Table 4: Reward mean vs strict success rate on 1.5B-WebShop, where partial credit makes the two metrics distinct (ALFWorld is binary, so SR coincides with reward mean). DUET’s +17.5pp advantage in strict SR also corresponds to a +0.065 advantage in reward mean over the strongest non-DUET baseline, indicating the gain is not driven by a small number of high-credit successes.

453 The ranking under reward mean matches the ranking under strict success rate, indicating DUET’s
 454 advantage on WebShop is not driven by a small number of high-credit successes.

455 C Scope of auxiliary signal-channel ablations

456 Table 2 in the main paper now reports ablations of all four components. Here we provide additional
 457 context on the BC and SC ablation results.

458 **BC (−13.5pp AF, −19.5pp WS).** BC serves as a cold-start safety net, providing dense token-level
459 imitation before the DR3 discriminator has accumulated enough data to be reliable. Its contribution
460 is consistent across both environments. Because BC and DR3 are coupled through the discriminator
461 (BC decays as discriminator accuracy rises), the BC ablation also partially disrupts the DR3 warmup
462 dynamic. The marginal contribution should therefore be interpreted as the value of having a cold-
463 start bridge, not purely the value of imitation.

464 **SC (−16.5pp AF, −35.0pp WS).** The State Channel provides dense exploration guidance through
465 progress-based reward shaping. Its larger contribution on WebShop (−35.0pp vs. −16.5pp on ALF-
466 World) reflects WebShop’s richer state space, where the progress map can provide more granular sig-
467 nal. On WebShop, removing SC drops performance more than removing DR3 (−35.0 vs. −26.5pp),
468 indicating that SC’s dense reward is the primary driver of exploration improvement on this environ-
469 ment, with DR3 providing complementary gradient correction.

470 D Extended analysis

471 This appendix provides additional analyses that complement the main experiments. All figures are
472 derived from existing training logs and saved trajectory data; no additional GPU experiments were
473 run.

474 D.1 State Channel dynamics

475 Figure 5 tracks three State Channel signals over training: SC progress (mean progress score on on-
476 policy trajectories), SC coverage (fraction of trajectory states matched in the expert progress map),
477 and bonus/reward ratio (SC bonus relative to environment reward). Across all four settings, SC
478 progress and coverage increase over training as the on-policy agent visits more expert-like states.
479 The bonus/reward ratio decreases as the environment reward grows: SC provides dense signal when
480 it is most needed (early training) and recedes as the real reward becomes informative.

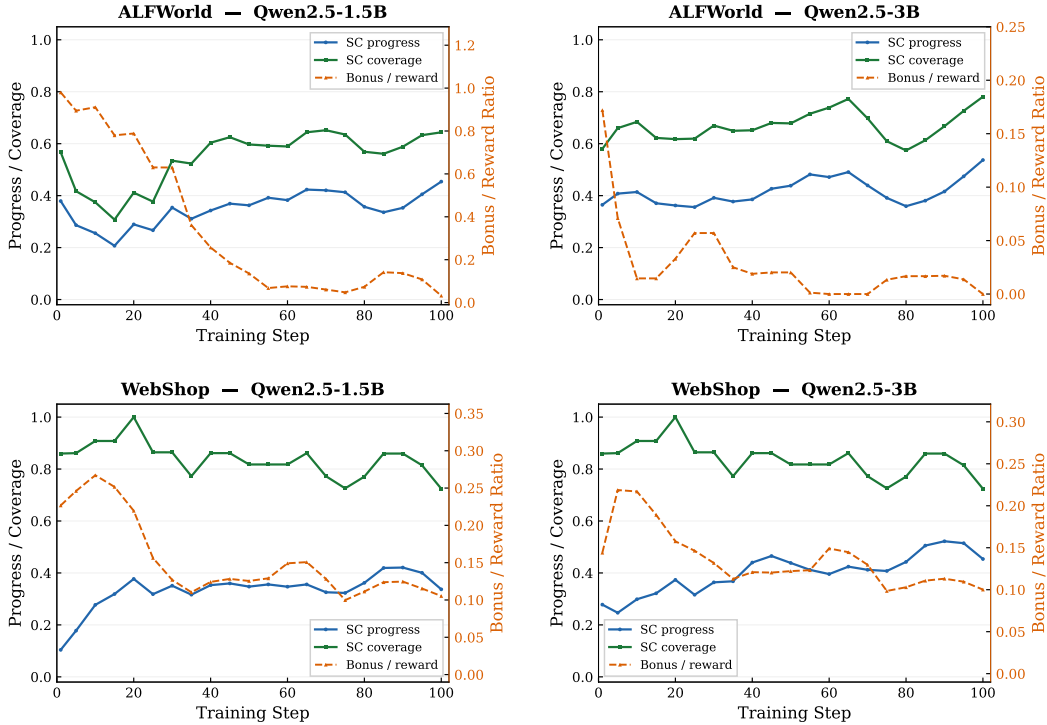


Figure 5: **State Channel dynamics.** SC progress (blue) and coverage (green) increase as the agent improves; the bonus/reward ratio (orange, right axis) decreases as environment reward grows.

481 D.2 Reward decomposition

482 Figure 6 decomposes the mean total reward for on-policy DUET trajectories into three components:
 483 environment reward, SC trajectory-level bonus, and SC step-level deltas. In early training, the SC
 484 components provide the majority of the reward signal (especially visible on ALFWorld 1.5B and
 485 WebShop). As training progresses, environment reward dominates, confirming that SC reward shap-
 486 ing provides a useful cold-start signal without distorting the converged policy’s reward landscape.

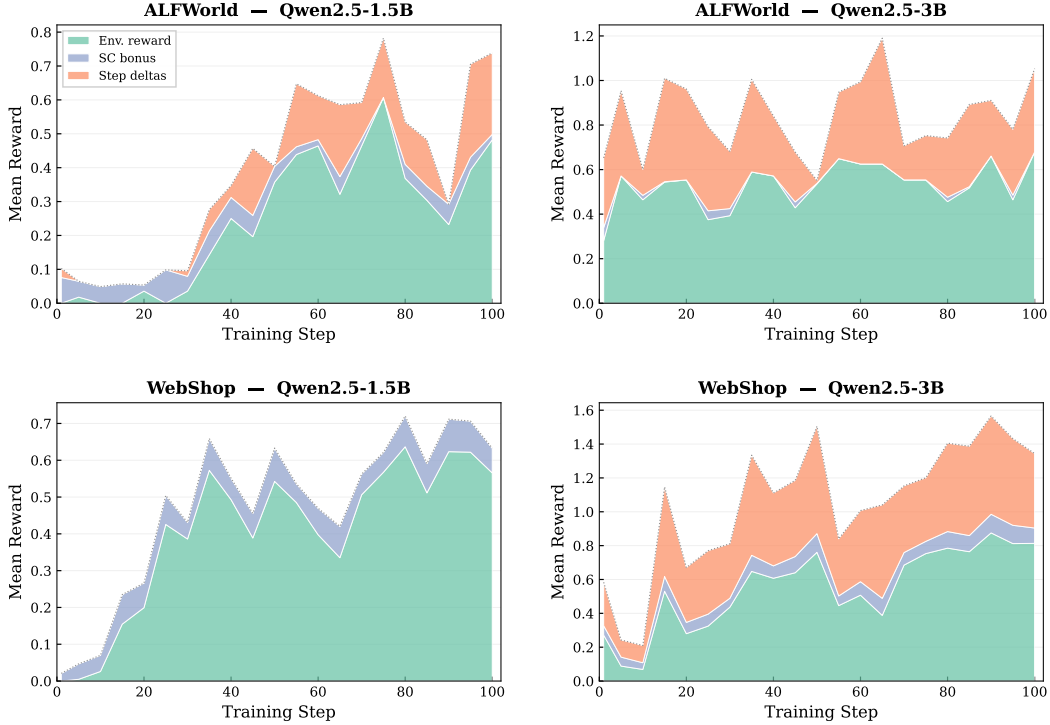


Figure 6: **Reward decomposition** over training. Green: environment reward. Blue: SC bonus. Orange: step deltas. SC components dominate early, environment reward dominates late.

487 D.3 Entropy comparison across methods

488 Figure 7 compares on-policy token entropy across all methods. DUET generally maintains higher
 489 or competitive entropy compared to baselines, suggesting that its dual-channel design preserves
 490 exploration diversity. This is most visible on ALFWorld 1.5B, where GRPO’s entropy collapses
 491 to 0.02 while DUET sustains 0.12—DUET’s dense SC reward signal provides sufficient learning
 492 gradient without requiring the policy to collapse its exploration.

493 D.4 Per-task-type breakdown (ALFWorld)

494 ALFWorld comprises six task types requiring different manipulation chains. Figure 8 decomposes
 495 success rate by task type for each method. At 1.5B, DUET leads on all six types, with the largest
 496 margins on multi-step tasks (Clean: 52% vs. 13% for CHORD; Cool: 29% vs. 17%). These tasks
 497 require chains of intermediate actions (e.g., find → pick up → go to sink → clean → go to destination
 498 → put), where the State Channel’s step-level progress bonuses are most informative. At 3B, the
 499 gap narrows as all teacher-mixing methods benefit from stronger base capabilities, but DUET still
 500 leads overall. One notable exception is Pick Two at 3B, where GRPO outperforms all teacher-
 501 mixing methods (51% vs. DUET’s 38%), suggesting that teacher demonstrations may not always be
 502 beneficial for tasks requiring broad combinatorial exploration.

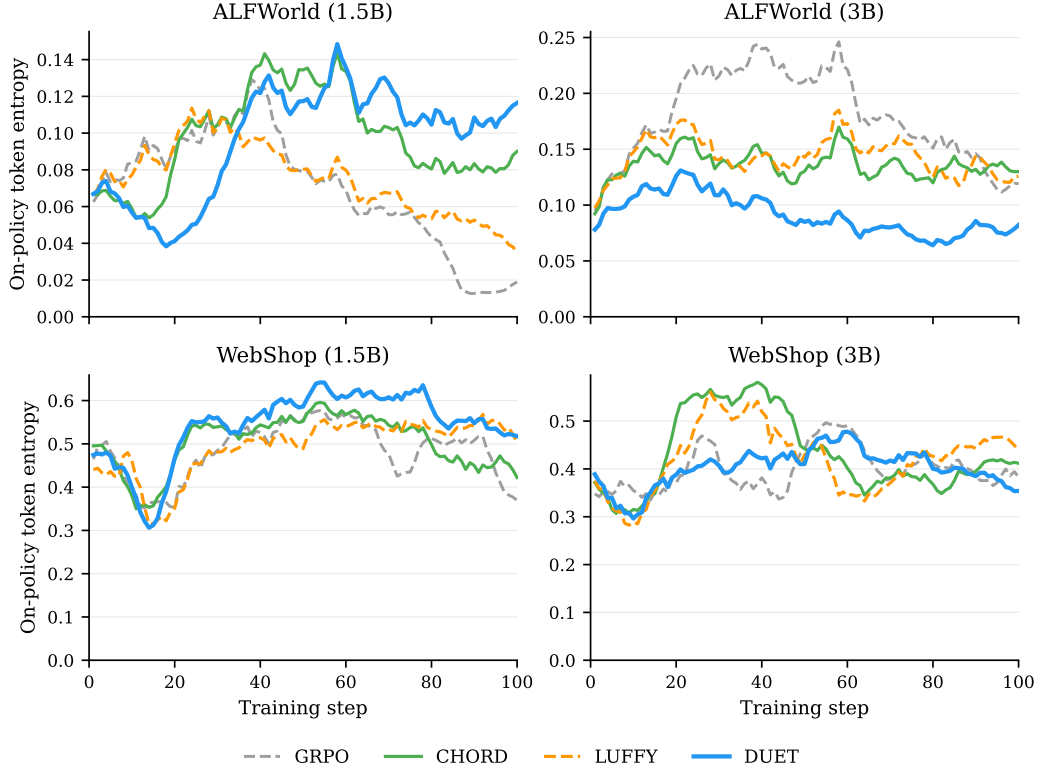


Figure 7: **On-policy token entropy** over training. DUET maintains higher entropy than GRPO in most settings, suggesting better preservation of exploration diversity.

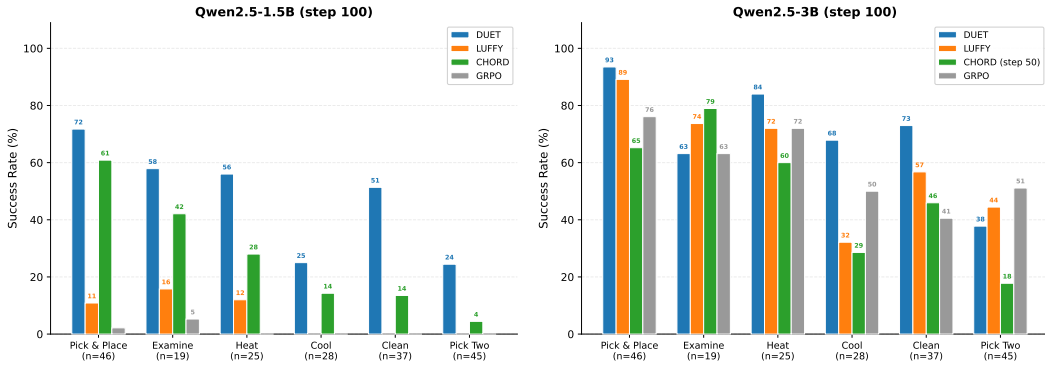


Figure 8: **Per-task-type success rate** on ALFWorld. DUET leads across all task types at 1.5B, with the largest gains on multi-step manipulation tasks (Cool, Clean).

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